

Answer **all** questions.

- 1** The Sunda slow loris (*Nycticebus coucang*) is a primate native to forests of Indonesia, western Malaysia, southern Thailand and Singapore. Listed as “Vulnerable” on the International Union for Conservation of Nature (IUCN) Red List, Its population is fast declining due to extensive habitat loss and illegal poaching for pet trade.

In order to understand more about the species before it is extinct, molecular research work is conducted by a wildlife conservation research facility. A gene of interest from the species and a bacterial plasmid vector are cleaved using the same restriction enzyme and ligated before being cloned in a bacterial host model organism for further analysis. The restriction enzyme used in the experiment generates sticky ends while the bacterial plasmid used contains marker genes.

(a) Explain how restriction enzymes cut DNA. [1]

- Each restriction enzyme recognizes and binds to a specific sequence of 4 to 8 nucleotide bases on the DNA molecule called a restriction site to hydrolyse the phosphodiester bond between two specific nucleotides in the restriction site.

(b) Explain what is meant by marker gene. [1]

- Gene which will give a visible phenotype / traits /characteristic to the bacteria that is easily detected for identification/selection of bacteria with vector/plasmid (**R.** recombinant vector/ plasmid);

(c) Suggest why it is possible to obtain two different types of recombinant plasmids in the experiment. [2]

- The restriction sites recognised by most restriction enzymes are palindromic/ each strand has the same sequence of bases when read from 5' to 3' end;
- Sticky ends of both ends of the gene of interest are the same if the fragment is rotated in either orientation/ gene of interest could be inserted in two different/ either orientation in the plasmid.

The Sunda slow loris is protected by law under the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) signed by many countries. It is an offence to be in possession or to trade in any illegally imported or acquired CITES species.

Before the Sunda slow loris is sold as pet, its teeth are often removed to prevent injury to potential buyers. However, this frequently results in infection and death of the animal. In a recent search of the home of an alleged wildlife trader, no live animal was recovered except three removed teeth of suspected slow loris species.

DNA from dental pulp extracted from these teeth were analysed based on several Variable Number Tandem Repeat (VNTR) loci. Fig. 1.1 shows the genetic fingerprints of the three teeth found, labelled as Teeth 1, Teeth 2 and Teeth 3, and teeth known to be taken from a Sunda slow loris.

DNA ladder	Teeth from Sunda slow loris	Teeth 1	Teeth 2	Teeth 3
6 kb	_____		_____	
5 kb	_____			
4 kb				_____
3 kb	_____	_____	_____	_____
2 kb	_____		_____	
1 kb		_____		_____

Fig. 1.1

(d) (i) Briefly explain the basis of genetic fingerprinting. [2]

- Different species/ individuals have different VNTR, hence differ in distance between restriction sites of a particular RE and there are different number of restriction sites due to restriction fragment length polymorphism (RFLP) ;
- The length and number of restriction fragments produced differ when the two DNA are digested with the same restriction enzyme .

- (ii) With reference to Fig. 1.1, which tooth is most likely removed from the Sunda slow loris? Explain your answer. [2]

- Teeth 2;
- its genetic fingerprint has three restriction fragments of length 2 kb, 3 kb and 6 kb that match restriction fragments from tooth sample of a known Sunda slow loris;
OR
genetic fingerprints of Teeth 1 and 3 have restriction fragments of 1 kb and 4 kb that do not match restriction fragments from tooth sample of a known Sunda slow loris.

Note: Difference at 5 kb restriction fragment between genetic fingerprints of Teeth 2 and known Sunda slow loris could be attributed to RFLP explained in (d)(i).

- (iii) Rescued captive Sunda slow lorises that have their teeth removed could not be reintroduced into the wild, as they are unlikely to survive. Attempts at breeding Sunda slow lorises in captivity in zoos were made. However, animals continuously bred in captivity might experience inbreeding depression.

Using your biological knowledge on restriction fragment length polymorphism, explain how the banding patterns of a genetic fingerprint can show whether or not an animal is inbred. [3]

- Compare banding pattern with other individuals of the same species. Differences in RFLP markers give bands in different places due to differences in length of restriction fragment produced;
- Common bands indicate related individuals;
- If the Sunda slow lorises are inbred, there would be a reduced number of bands between the Sunda slow lorises compared. More inbred, the greater the similarities in banding patterns.

[Total: 11]

- 2 Cystic fibrosis (CF) is one of the most common genetic disorders in the United States. It is estimated that 30,000 Americans are living with CF and approximately 1,000 new cases are diagnosed every year. CF is caused by mutations in a gene located on chromosome 7.

(a) Explain how CF is inherited.

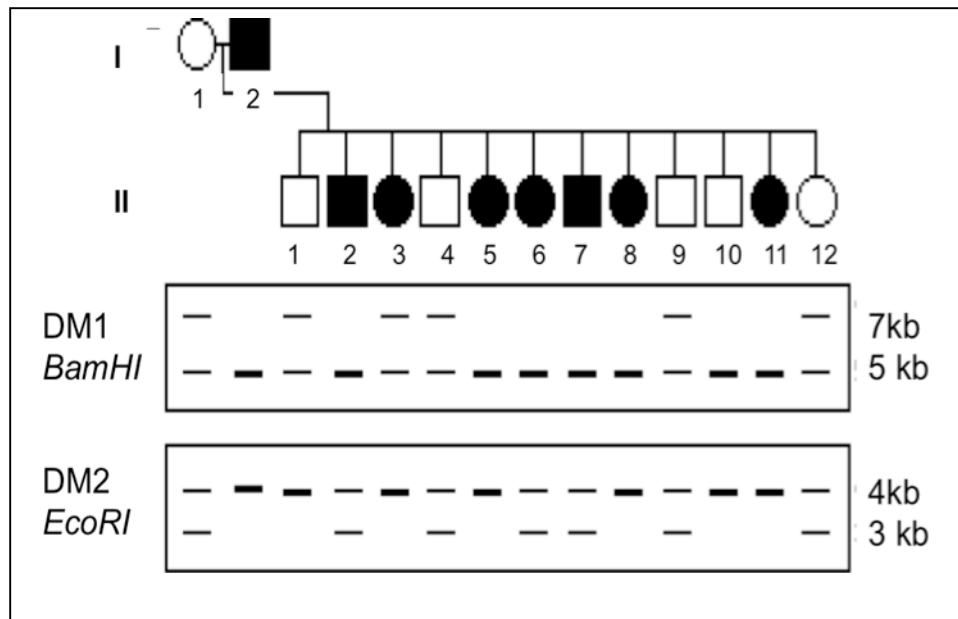
[2]

- autosomal recessive condition;
- *autosomal* – males and females are equally affected/ expression of characteristic not dependent on the sex of the individual;
- *recessive* – requires both alleles to be defective before the child is phenotypically CF; (*Reject: individual must be homozygous recessive because recessive is not explained*)
- carrier x carrier – $\frac{1}{4}$ chance of producing a CF child;

[max 2]

In the early 1980s, attempts were made to map the CF gene on chromosome 7 using DNA markers.

The relationship between the CF gene and two DNA markers DM1 and DM2 were studied in a family with many affected children. DNA were extracted from these individuals and cut with two different restriction enzymes, *Bam*HI and *Eco*RI, separately. The products of digestion were separated using gel electrophoresis and the RFLP banding patterns is shown in Fig. 2.1 below.



(Note: *Bam*HI was used on DM1 and *Eco*RI on DM2)

Fig. 2.1

- (b) (i) Using the information provided in Fig. 2.1, deduce which marker is more suitable in detecting the presence of cystic fibrosis? [3]

- DM1 – more suitable /DM2 – less suitable;
(mark awarded only if the student is able to provide some support for the answer.)

Either

- Becos for DM1, 7 out of 8 (88%) of the CF individuals have the **only one 5kb band/ homozygous for the 5 kb fragments**;
- 5 out of 8 (62.5%) of the CF individuals have **only one 4kb band/ homozygous for the 4 kb fragments** for DM2;

Or

- For DM1 - All CF individuals have **only one 5kb band/ homozygous for the 5 kb fragments** except II-3 who has 2 bands 5kb and 7 kb;
- For DM2 - 5 CF individuals have **only one 4kb band/ homozygous for the 4 kb fragments** and 4 CF individuals have 2 bands 3kb and 4kb / about equal number of CF individuals and normal individuals have a single 4kb band as well as 2 bands 3kb and 4kb;
(So the inheritance of the DM1/5kb fragment is more consistent compared to DM2)

- (ii) Using your understanding of linked genes, provide an explanation for your answer to (b)(i). [2]

- DM1 **closer** to CF gene compared to DM2;
(note: some form of comparison required between DM1 and DM2 before a full mark is given.)
- Higher tendency for the 5kb fragment to be inherited together with the CF gene;
- Hence lower number of recombinants (individuals II-3 and II-10 compared to DM2 (II-1,2,,6,7 and 10);

[max 2]

- (c) An experiment using a non-viral vector to deliver the cystic fibrosis transmembrane conductance regulator (CFTR) gene was carried out on five individuals with cystic fibrosis (CF). Three doses of plasmids bearing the CFTR gene were administered to the patients at four-week intervals. After each dose, tests were carried out on the patients to detect the following: plasmid-derived CFTR DNA, plasmid-derived CFTR mRNA, plasmid-derived CFTR protein and CFTR protein function. The results are shown in Table 2.2, with positive samples marked '+’.

Table 2.2

Patient	Dose 1				Dose 2				Dose 3			
	DNA	mRNA	protein	protein function	DNA	mRNA	protein	protein function	DNA	mRNA	protein	protein function
1	+	+	+	+			+		+			
2									+	+	+	+
3												
4	+	+	+		+	+	+					
5	+	+			+	+			+	+		

DNA : detection of plasmid-derived CFTR DNA
 mRNA : detection of plasmid-derived mRNA
 protein : detection of plasmid-derived CFTR protein
 protein function : detection of CFTR protein function

For (c) (i)-(iv), use the data from Table 2.2.

- (i) Calculate the percentage success rate of CFTR gene delivery by liposomes for all 15 doses. [1]
- $8/15 = 53.3\%$
- (ii) Calculate the percentage success rate of transcription of plasmid-derived CFTR DNA for cases where gene delivery was successful. [1]
- $7/8 = 87.5\%$
- (iii) From your answer in (c)(i) and (c)(ii), determine whether gene delivery or transcription of the delivered gene was the more significant obstacle to successful gene therapy. [1]
- gene delivery

- (iv) Suggest an alternative delivery method that can help address the obstacle discussed in (c)(iii). [1]
- Use adenovirus as vector for in vivo inhalation of aerosol spray
- (d) Currently, gene therapy involves the introduction of the normal allele into somatic cells of individuals suffering from CF.
- (i) Explain why there seems to be fewer objections to this method compared to one where gene therapy is applied to gametic cells. [2]
- Somatic cell gene therapy: (1 max)
- Only the individual receiving the treatment is affected;
 - The allele that is introduced is not inherited by his offspring;
 - Individual has consented to the therapy;
- Gametic cell gene therapy: (1 max)
- Inheritable by the offspring/ future generations;
 - Such treatment produces generations of unconsenting research subjects;
 - Long term effects of gene therapy still unknown;
 - May tamper with human evolution eg. genes which cause harm in one environment may be of evolutionary significance in another;
- (ii) Explain why gene therapy provides a possible cure for CF but not stem cell therapy. [2]
- Multiple organs affected – lungs, pancreas etc. / many different cell types are involved;
 - not all organs involved have stem cells that can be used for stem cell therapy/ difficulty in extracting the stem cells for use; Difficulty in obtaining stem cells that are able to differentiate into many different types of cells;

[Total: 15]

- 3 A number of diseases, such as dengue fever, are spread by mosquitoes. The incidence of this disease has increased dramatically in recent years and this has been linked with the spread of the mosquito, *Aedes aegypti*.

In an attempt to reduce the numbers of *A. aegypti*, genetically modified (GM) male mosquitoes were produced. One of the genes added to these mosquitoes, **when switched on**, results in the production of a protein that is toxic to mosquitoes.

In 2010, in the Cayman Islands and in Malaysia, GM male mosquitoes were released into the wild to mate with females. All the resulting offspring died in the larval stage.

In Malaysia, both GM male and non-GM male mosquitoes were released in order to compare their dispersal and life span in the wild. The GM mosquitoes could be identified because they also carried a gene for green fluorescent protein (GFP).

- (a) Explain why, in many examples of gene technology, fluorescent markers are used in preference to antibiotic resistance genes. [1]

- easier to, identify/screen for transformed cells;
- more economical/time saving/labour saving;
- resistance gene(s) can be passed to other bacteria;
- idea of antibiotics no longer effective

OR

requiring development of new, antibiotics/treatments;

[max 1]

- (b) In addition to the gene for GFP, the DNA that has been added to the GM mosquitoes consists of

- a promoter
- a gene coding for a toxic protein, tTA
- a binding site for tTA.

When a GM mosquito larva hatches from an egg, the promoter induces the production of only a small amount of tTA, so that the larva does not die immediately. In a process of positive feedback, the tTA produced binds to the DNA as shown in Fig. 3.1. This increases the expression of the gene until the increased concentration of tTA kills the larva.

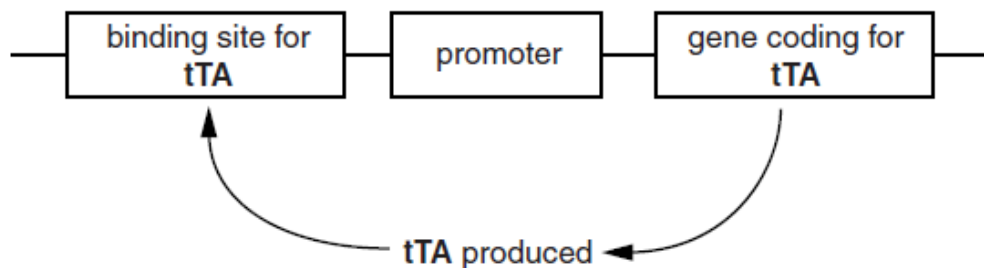


Fig. 3.1

- (i) Explain why, in gene technology, a promoter needs to be transferred along with the desired gene. [2]

- RNA polymerase and general transcription factors bind at the promoter to initiate transcription / switch on gene expression;
- If the promoter is not transferred along with the desired gene, the gene has to be inserted near an existing promoter, this may disrupt expression of existing gene;
- in eukaryotes precise position of promoter is important for gene expression;

[max 2]

- (ii) Switching on the gene coding for tTA in the mosquito larvae, rather than in the eggs, increases the effectiveness of this method of controlling mosquito numbers.

Explain why this is so.

[2]

- GM larvae do not die immediately thus this gives longer time for tTA production / tTA to build up;
- so tTA can get into environment (when GM larvae die) and kills non-GM larvae;

- (c) GM mosquitoes carrying the tTA gene can live and reproduce normally when fed on a diet containing an added chemical, **A**.

With reference to Fig. 3.1,

- (i) suggest how A could prevent death of the GM mosquitoes. [1]

- chemical A has, similar shape to tTA/ complementary shape to DNA binding site, thus chemical A can bind to the binding site, and prevents tTA from binding;

OR

- chemical A, binds to/changes shape of tTA, thus prevents tTA from binding to DNA binding site, this stops the positive feedback/ small quantity of tTA does not kill;

OR

- chemical A, binds to/changes shape of/breaks down of tTA, thus tTA is no longer toxic;

[Any 1, max 1]

- (ii) suggest why there is little danger of the gene carried by these GM mosquitoes being passed to other organisms from GM mosquitoes which escape or are released into the wild. [1]

- GM males die if they cannot get chemical A;
- (if males mate), their offspring die;
- GM males only mate with other *A. aegypti* / their own species;

[Any 1, max 1]

In 1907, scientists found that a common plant tumour was caused by the invasion of a bacterium called *Agrobacterium tumefaciens*. They discovered that the tumour was caused when plasmids from the bacteria were taken into the DNA of the host plant's cells and expressed. By 1983, biotechnologists were able to introduce genes into plant cells using modified plasmids that do not result in tumour formation. Fig. 3.2 shows the procedure of transforming *Brassica* using this method.

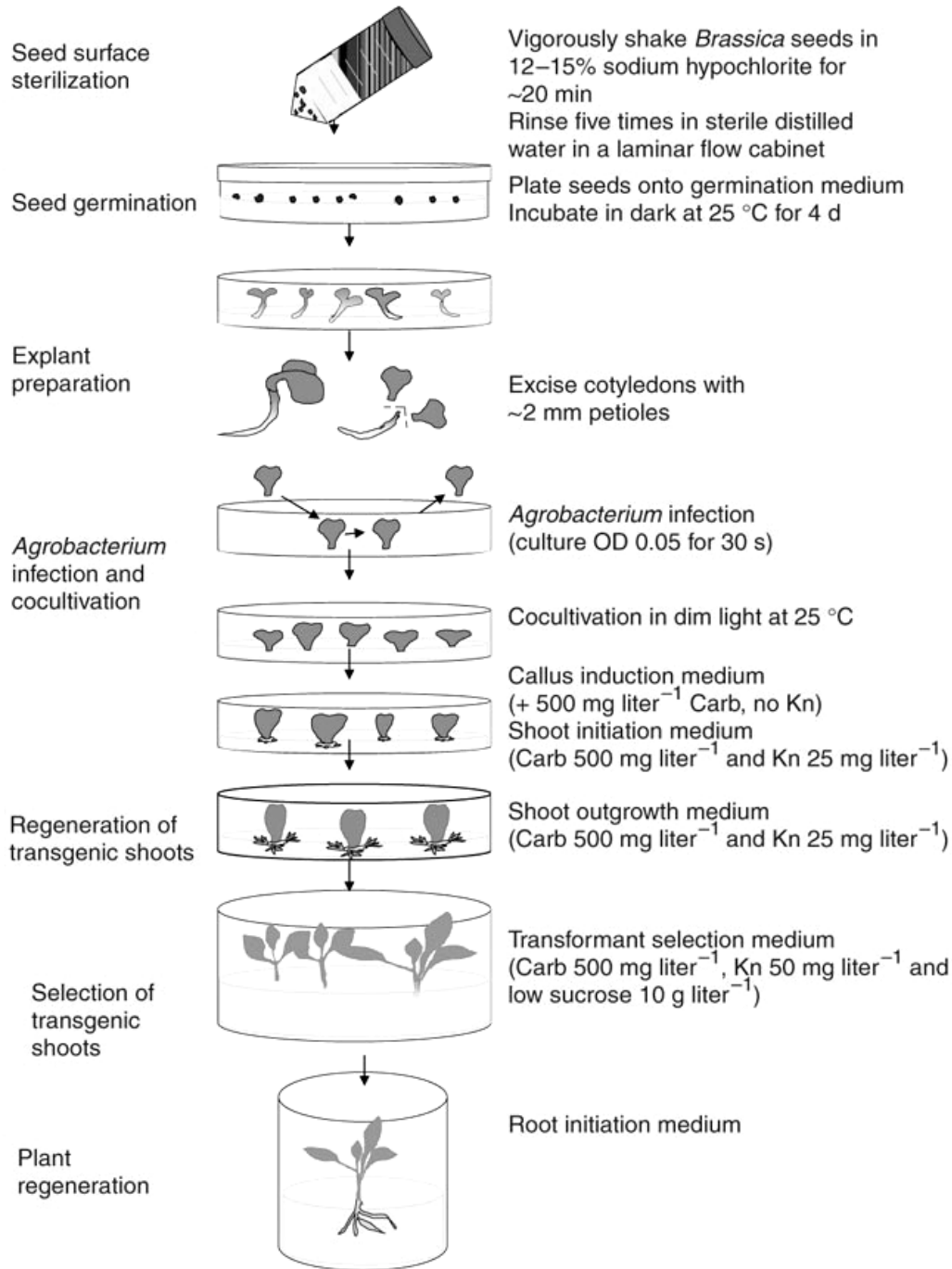


Fig. 3.2

(d) With reference to Fig. 3.2, explain why carbenicillin and kanamycin were added in the various media. [3]

- Carbenicillin and kanamycin are antibiotics, which were added to select for the growth of transformed Brassica;
- OR
- Carbenicillin and kanamycin are antibiotics; carbenicillin is added to inhibit the growth of *A. tumefaciens* attached to the explants and kanamycin added subsequently to select for the growth of transformed Brassica;
- The genetic markers on the plasmid are therefore genes resistant to carbenicillin and kanamycin;
- Only successfully transformed cells/ plantlets will survive in culture;

(e) Explain why plant tissue culture technique is used to regenerate the transformed plants. [2]

[Note: Question focuses only on the final stage of “plant regeneration”]

(Any 2:)

- They have the advantage of being free from fungal or bacterial infections;
- Quick production of mature plants;
- Regeneration of whole plants from genetically modified explants;
- Amount of space required for field trials to determine which plants have desirable traits is reduced;

(f) State the difference in composition between the root initiation medium and the shoot outgrowth medium. [1]

- Root initiation medium: high auxin:cytokinin whereas;
Shoot outgrowth medium: high cytokinin:auxin

(g) Suggest why seeds were used instead of explants from adult *Brassica* plants, as starting plant material for this procedure. [1]

- Seeds are easier to surface sterilise than adult plant parts/ more effective surface sterilisation

OR

- Adult plants may be diseased, thus lowering the chances of successful cultures

OR

- Any other valid point;

[Total: 14]

4 **Planning question**

Content	Expected answer	Marks
P Defining the problem	P: Planning 1. Hypothesis or prediction e.g. tolerant variety has a lower (more negative) solute potential than non-tolerant / incipient plasmolysis brought about in tolerant variety by higher concentration of external solution than in case of cells of the non-tolerant variety;	[max 2]
	2. Theory to support hypothesis or prediction water moves by osmosis from, higher to lower / towards the more negative solute potential therefore when external solute potential is low (as in a concentrated salt solution) the internal solute potential must be even lower (as in more concentrated internal solution)	
	3. Outline of strategy and justification e.g. determining the external concentration / solute potential which results in incipient plasmolysis / 50% plasmolysis in cells of each variety <u>Key variables:</u>	[max 1]
	4. independent variable: concentration of / solute potential of, external solution; 5. dependent variable: number / percentage, of cells plasmolysed; 6. controlled variable: immersion time and volume of immersion solution;	
	7. Risk Assessment: must refer to hazard and precaution e.g. care in attaching pipette filler to pipette / care in use of scalpel to remove epidermal tissue from onion / dispose of broken cover slips with care Acceptable: Some points may be taken from a diagram or a flow sequence diagram.	[max 1]

Content	Expected answer	Marks
M Methods	<u>S: Preparation of solutions</u> 8. prepare range of sodium chloride concentrations using the 1 mol dm ³ solution provided ; 9. suggested range of 0 and at least 5 concentrations at equal intervals, e.g. 0.2, 0.4, 0.6, 0.8, 1.0 mol dm ³ ; 10. how dilution made: e.g. to make 100 cm ³ of 0.2 mol dm ³ solution, pipette 10 cm ³ 1 mol dm ³ solution into 100 cm ³ volumetric flask or measuring cylinder and make up to 100 cm ³ with deionised water; 11. evidence of awareness that it is easier to make accurate dilution using an appropriate sized pipette and reasonably large total volume;	[max 2]
M Methods	<u>O: Obtaining results</u> 12. remove epidermis from inner surface of onion scale with scalpel; 13. and immerse in constant volume of solution; 14. in labelled specimen tubes; 15. leave for minimum of 15 minutes; 16. ref. to staggering to allow for sampling; 17. (after time period) transfer specimen to microscope slide; 18. in drop of external solution and add coverslip; 19. may add iodine as a coloured liquid – make plasmolysed cells easier to see; 20. (for each specimen) count cells and record % plasmolysed; 21. repeats at each concentration for both varieties of onion;	[max 2]
	<u>Analysis of results</u> G: Graph 22. plot graph concentration of external solution (x-axis); 23. against mean % plasmolysis (y-axis); 24. draw line of best fit (probably sigmoid); 25. plot both varieties of onion separately (could be on same axes); 26. read concentration causing 50% plasmolysis for each variety;	[max 2]
	<u>Analysis of results</u> Biostatistics: Mathematical processing 27. then the difference could be tested for significance using a biostatistical test / t-test; 28. in making the conclusion the solute potential of the cells of each variety of onion is equivalent to that of the solution which causes 50% plasmolysis; 29. because at this point, wall / turgor, pressure is zero therefore external and internal solute potential are equal; 30. the solute potential of a given concentration of sodium chloride solution can be found by consulting an appropriate table; 31. solute potential can be expressed in kPa and is normally a negative value;	[max 2]

[Total: 12]

Free-response question

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where applicable.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 (a)** Outline how a named protein can be produced by genetic engineering for the treatment of diabetes mellitus. [6]

1. Insulin can be produced for the treatment of diabetes mellitus;
2. Based on known amino acid sequences;
3. DNA constructs of A chain and B chain of human insulin were synthesised;
Reject: Gene of A chain
4. Each gene was placed under the control of a *lac* promoter;
5. And part of the β -galactosidase structural gene;
6. Both recombinant plasmids;
7. Were transformed separately into *E. coli*;
8. Transcription of *lac* promoter switched on when *E. coli* were grown in the presence of lactose;
9. The 2 artificial genes were expressed independently as fusion proteins;
10. The correct signal for initiation of translation is provided by the region immediately upstream of the β -galactosidase gene;
11. Cyanogen bromide is used to cleave the methionine residue;
12. Separating insulin polypeptides from the β -galactosidase fragments;
13. Purified A and B chains are attached to each other via disulphide bonds to form functional insulin;

[max 6]

- (b) Discuss the benefits and associated ethical implications of the human genome project. [8]

Benefits	Ethical implications
▪ Pharmacogenomics / knowing about which genes affect a person's response to a drug / genetic differences affect the way we react to a drug. ▪ Possibility of tailoring drugs prescribed to fit patient's pharmacogenomic profile for greater efficacy / avoiding dangerous side effects.	▪ Most effective drug may not be affordable for patient. Such testing may not be affordable & psychological impact of knowing treatment is available but not affordable.
▪ Risk assessment for certain diseases. Knowledge of genetic variations can help prevent / treat these diseases.	▪ Risk profiles for diseases may be accessible to insurers, employers, schools etc. / individual's privacy infringed. May be used in discriminatory manner/ stigmatization/ social risks. ▪ Psychological impact due to an individual's genetic differences. Doctors, other health service providers, patients and the public must be educated to make informed choices, and be aware of scientific capabilities and limitations. ▪ Standards and quality-control measures in testing procedures must be enforced to ensure reliability of tests. Proper counselling must be done in conjunction with risk profiling and where there is no treatment for certain diseases. ▪ Reproductive issues / decision making / rights and potentially controversial procedures e.g. eugenics, foetal genetic testing and abortion.
▪ Insights via comparative genomics / comparing human genome sequence and the genome sequence of model organisms / identification of genetic similarities and differences to model organisms. ▪ Leads to gene discovery / follow disease development / study disease treatment / elucidating evolutionary history and relationships between genes, genome and species.	▪ Infringement of animal rights. Also, animal disease models may not accurately reflect disease in humans.

Benefits	Ethical implications
▪ Large scale studies on genetically homogeneous populations to identify alleles linked to certain disease conditions.	▪ Uncertainties associated with gene tests. ▪ Genetic information of individuals becoming private intellectual property of biotechnology companies. Commercialization of related technologies derived from such research, but possibly no reciprocal direct benefit for individuals contributing their genetic information to the research.
[1mark per bulleted point, max 8]	

(c) Describe and give examples of unique features of zygotic stem cells, [6] embryonic stem cells and adult stem cells.

1. Zygotic stem cells [Example: Fertilised zygote]

- These are the cells in morula, which are formed during the first few divisions after fertilisation/fusion of an egg and sperm.
- Zygotic stem cells, which include the zygote and early embryo cells, are totipotent.
- They have the ability to differentiate into any cell type to form whole organisms and hence are also pluripotent and multipotent.

Functions :

- Zygotic stem cell has the potential to differentiate into all the cells that form the complete organism plus the extra-embryonic tissues which include the placenta and amnion.
- These cells give rise to form all cells in the body of the organism therefore they have limited ability to self-renew.

2. Embryonic stem cells

- Embryonic stem cells are pluripotent.
- They have the ability to differentiate into almost any cell type to form any organ or type of cell and so are not totipotent but are multi-potent.
- They are obtained from the Inner Cell Mass on the fourth day of development where the embryo forms into two layers, an outer layer (extra-embryonic membranes) and an inner cell mass (ICM).

○ Functions :

- Outer layer forms the placenta.
- Inner cell mass (ICM) forms the three primary germ layers: ectoderm, endoderm and mesoderm, which later form all the tissues of the human body.
- give rise to various organs in organism / multiple

specialized cell types that make up the heart, lung, skin and other tissues in the developing foetus.

3. Adult stem cells [E.g. Blood stem cells, Stromal Stem cells, Gut Epithelium stem cells]
- Blood stem cells are multipotent. / Found in each tissues or organs which help to replenish worn-out or damaged cells.
 - They have the ability to differentiate into a limited range of cell type and so are not pluripotent or totipotent.
 - Found in bone marrow or blood/skin etc
 - Functions:
 - Responsible for constant renewal of blood.
 - Give rise to all the blood cell types

[max 2 marks per type of stem cells, total 6 max]